2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
–1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum(x_i-\bar{x}) = 0$	$\sum(x_i-\bar{x})^2 = 10$	$\sum(y_i-\bar{y}) = 0$	$\sum(x_i-\bar{x})(y_i-\bar{y}) = 8$

(a) $\bar{x} = 1, \ \bar{y} = 2$

(b) $b_2 = \dfrac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \dfrac{8}{10} = 0.8$

$b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8 \times 1 = 1.2$

(c) $\sum_{i=1}^{5} x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$

$\sum_{i=1}^{5} x_i y_i = 3\times4 + 2\times2 + 1\times3 + (-1)\times1 + 0\times0 = 18$

$\sum_{i=1}^{5} x_i^2 - N\bar{x}^2 = 15 - 5\times1^2 = 10 = \sum_{i=1}^{5}(x_i-\bar{x})^2$

$\sum_{i=1}^{5} x_i y_i - N\bar{x}\bar{y} = 18 - 5\times1\times2 = 8 = \sum_{i=1}^{5}(x_i-\bar{x})(y_i-\bar{y})$

(d) $s_y^2 = \dfrac{10}{4} = 2.5 \quad s_{xy} = \dfrac{8}{4} = 2$

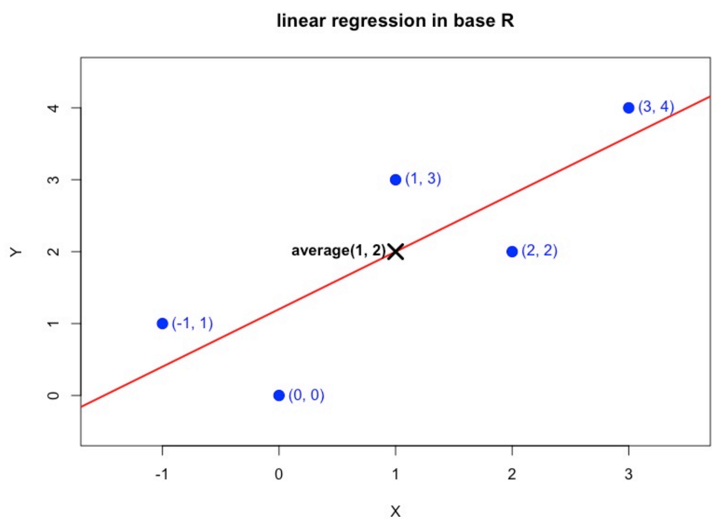
$s_x^2 = \dfrac{10}{4} = 2.5 \quad r_{xy} = \dfrac{s_{xy}}{s_x s_y} = \dfrac{2}{\sqrt{2.5}\sqrt{2.5}} = 0.8$

$CV_x = 100\left(\dfrac{s_x}{\bar{x}}\right) = \dfrac{100\sqrt{2.5}}{1} = 50\sqrt{10}$

median $(x) = 1$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	−0.8	0.64	−1.6
1	3	2	1	1	1
−1	1	0.4	0.6	0.36	−0.6
0	0	1.2	−1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i\hat{e}_i = 0$

(e) (f)



(g) $b_1 = 1.2$

$b_2 = 0.8$

$\bar{x} = 1$

$\bar{y} = 2$

$\bar{y} = b_1 + b_2\bar{x} \Rightarrow \bar{y} = 2 = b_1 + b_2\bar{x} = 1.2 + 0.8(1) = 2$

(h) $\bar{\hat{y}} = \sum_{i=1}^{5} \frac{\hat{y}_i}{N} = \frac{3.6+2.8+2+0.4+1.2}{5} = 2 = \bar{y}$

(i) $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

(j) $var(b_2) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$se(b_2) = \sqrt{var(b_2)} = \sqrt{0.12} = \frac{\sqrt{3}}{5}$

2.22

(a)

```
Residuals:
    Min      1Q  Median      3Q     Max 
-189.44  -50.44   -9.44   43.06  324.38 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.675 249.396   <2e-16 ***
small         12.175      5.369   2.268   0.0236 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608,	Adjusted R-squared:  0.005323 
F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

$$TOTALSCORE = 916.442 + 12.175\ SMALL$$
$$y = \beta_1 + \beta_2 x$$

$\Rightarrow \beta_1$: Average total score for students in regular-sized classes.

β_2 : Students is small classes score on average 12.175 points higher than those in regular classes.

Conclusion : Small class sized have a positive impact on overall student achievement.

(b)

```
Residuals:
    Min      1Q  Median      3Q     Max 
-74.665 -21.590  -2.665  16.410 194.335 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.488 290.760  < 2e-16 ***
small          6.924      2.174   3.185  0.00151 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295,	Adjusted R-squared:  0.01167 
F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
Residuals:
     Min       1Q   Median       3Q      Max 
-144.777  -35.028   -5.028   29.223  142.223 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.449 197.545   <2e-16 ***
small          5.251      3.578   1.467    0.143    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778,	Adjusted R-squared:  0.001488 
F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

Reading : $\beta_{small} = 6.924$

Math : $\beta_{small} = 5.251$

Conclusion : The small-class effect is much stronger and more reliable for "Reading" than for "Math" in this specific kindergarten sample.

(c)

```
Residuals:
     Min       1Q   Median       3Q      Max
-191.748  -50.442   -5.442   40.558  312.558

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  916.442      3.559 257.529   <2e-16 ***
aide           4.306      4.994   0.862    0.389
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898,	Adjusted R-squared:  -0.0003068
F-statistic: 0.7436 on 1 and 835 DF,  p-value: 0.3888
```

$$\text{TOTALSCORE} = 916.442 + 4.306 \text{ AIDE}$$

$$y = \beta_1 + \beta_2 \quad x$$

$\Rightarrow$ β_1 : Average score in regular classes.

β_2 : Adding a teacher aide increase the score by only 4.306 points on average.

Conclusion : There is no strong evidence that adding a teacher aide improves total score.

(d)

```
Residuals:
    Min      1Q  Median      3Q     Max
-74.665 -21.536  -2.536  15.464 194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665      1.460 296.341   <2e-16 ***
aide           2.871      2.049   1.401    0.161
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared:  0.002347,	Adjusted R-squared:  0.001152
F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615
```

```
Residuals:
     Min       1Q   Median       3Q      Max
-146.212  -31.212   -5.777   29.223  142.223

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777      2.355 205.464   <2e-16 ***
aide           1.435      3.304   0.434    0.664
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared:  0.0002258,	Adjusted R-squared:  -0.0009715
F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

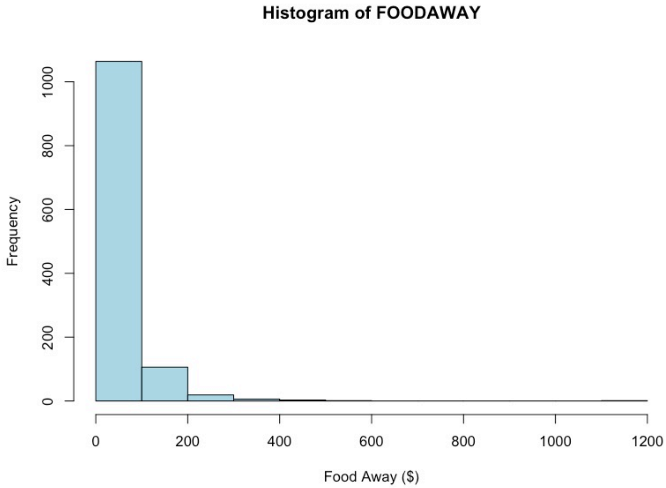
Reading : $\beta_{aide} = 2.871$ ($p = 0.161$), Not significant

Math : $\beta_{aide} = 1.435$ ($p = 0.664$), Not significant

Conclusion : Unlike small class sizes. adding a teacher aide does not show a significant benefit in either subject.

2.25

(a)



Mean = 49.27
Median = 32.55

25th percentiles : 12.04
75th percentiles : 67.5

(b)

advance :

1st Qu.	Median	Mean	3rd Qu.
21.67	48.15	73.15	90.00

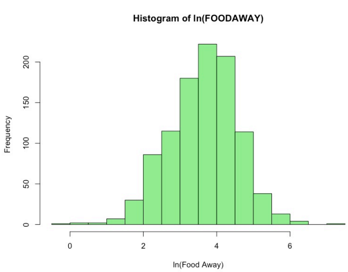
college :

1st Qu.	Median	Mean	3rd Qu.
14.44	36.11	48.60	68.67

no advance or college :

1st Qu.	Median	Mean	3rd Qu.
9.63	26.02	39.01	52.65

(c)



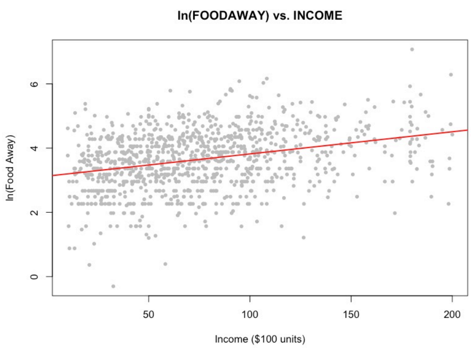
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.3011	3.0759	3.6865	3.6508	4.2797	7.0724

The reduction in observation for ln(FOODAWAY) is due to the exclusion of zero-expenditure households, as the natural log is only defined for positive value.

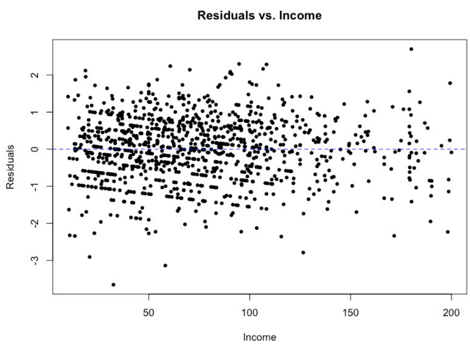
(d) $\ln(FOODAWAY) = 3.1293 + 0.0069017\, INCOME$

$\beta_2 = 0.0069$ means one-unit increase, the average expenditure on food expect increase 0.69%.

(e)



(f)



The residual plot shows a random scatter of points around the zero line with no distinct system. It means the model is complete random.